SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B.Tech/MBA.Tech

Stream/s : Data Science

Subject: MLOps

Date: 22/2/25

Marks: 100

Year: IV

Semester: VII

Time: 3 hrs (2:00pm to 5:00pm)

No. of Pages:3

Re-Examination (24-25)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right-hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1 SO-1 BL- 1	a.	Explain the concepts of Traditional Waterfall model as compared to Agile Methodology	[5]
CO-2 SO-2 BL- 2	b.	What are the key challenges arises in deploying ML models in production?	[5]
CO-3 SO-1 BL-2	c.	Explain the concept of Infrastructure as Code in brief	[5]
CO-3 SO-2 BL-1	d.	What are Data Distribution Shifts and Model Drift with respect to MLOPs?	[5]
Q2 CO-2 SO-3 BL-3	A	What is a CICD pipeline? Explain various stages involved in the MLOps pipeline.	[10]
Q2 CO-3 SO-3 BL-3	B	Explain the Core Components of MLflow with diagram	[10]

Q3 CO-1 SO-1 BL-1	A	Describe the significance of scrum for Agile methodology with respect to their Components (like Roles, Artifacts and Events) and Process Flow	[10]
Q3 CO-2 SO-2 BL-2	B	Discuss in brief the importance of source code control in machine learning projects, including its historical development with its role in version control and collaboration	[10]
Q4 CO-3 SO-2 BL-1	A	What is Kubernetes? Explain its architectural components in brief with diagram	[10]
Q4 CO-3; SO-3; BL-3	B	Write a short note on the following a) Airflow b) DVC c) Jenkins Pipeline d) AzureDevOPs	[10]
Q5 CO-3; SO-2; BL-1	A	What is Containerization? Explain the lifecycle on containers with its commands	[10]
Q5 CO-3; SO-2; BL-2	B	Discuss the various model packaging and deployment types, as well as the platforms to host the models.	[10]
Q6 CO-1 SO-1 BL-2	A	Explain the ML Project Lifecycle along with the pitfalls of focusing on only ML models?	[10]
Q6 CO-2 SO-2 BL-2	B	Compare between a) ETL Pipeline vs CICD Pipeine b) Data lakes vs. Data warehouses c) Batch processing vs Stream processing with respect to MLOPs d) DevOPs vs MLOPs	[10]
Q7 CO-3 SO-2	A	Describe Feature Stores with its architecture in detail.	[10]

BL-4			
Q7 CO-3 SO-3 BL-3	B	What is Model Retraining with respect to ML projects? Why it is required? Explain the four main approaches for retraining	[10]